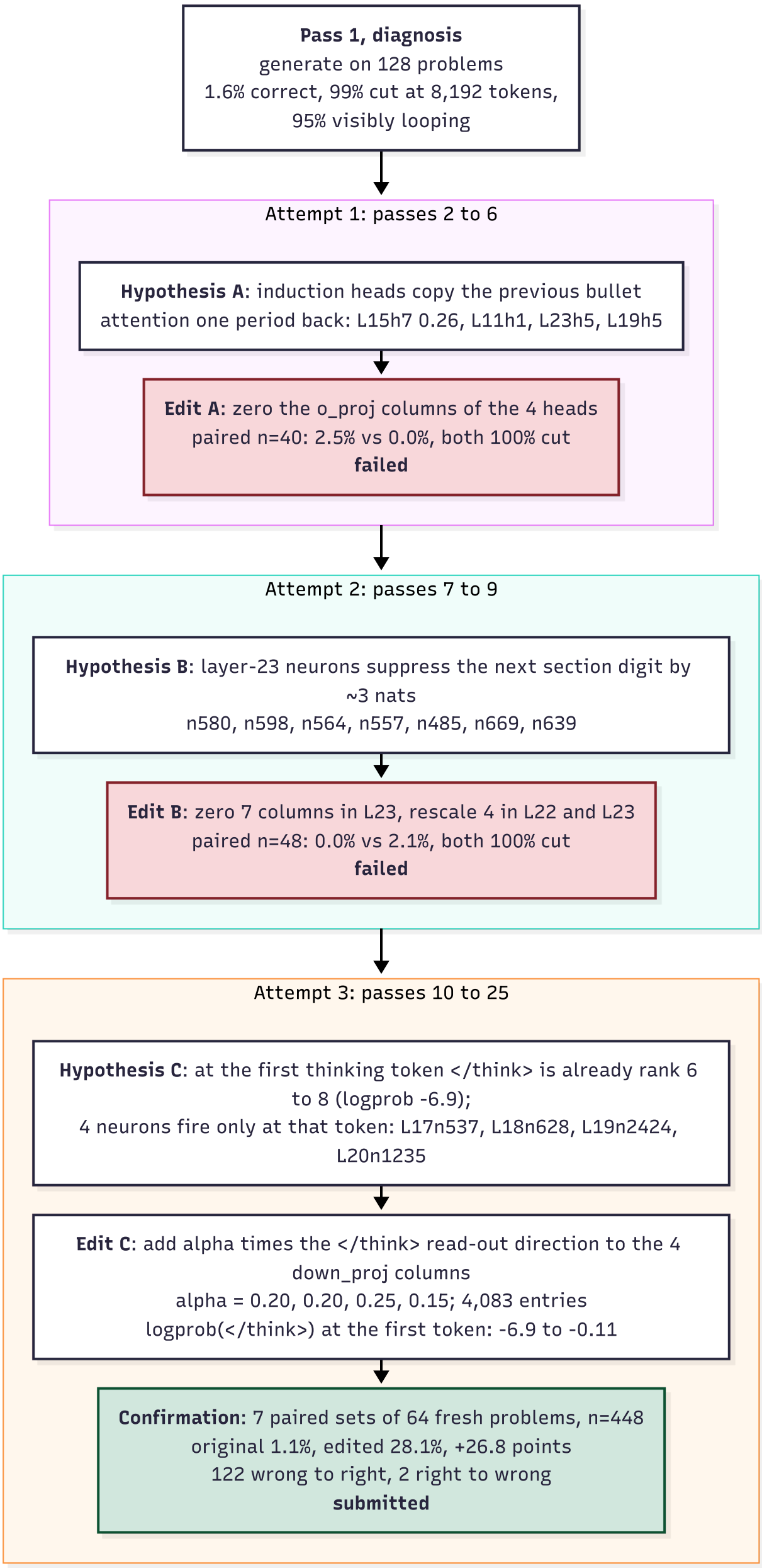

Pass 1, diagnosis
generate on 128 problems
1.6% correct, 99% cut at 8,192 tokens,
95% visibly looping
Attempt 1: passes 2 to 6
Hypothesis A: induction heads copy the previous bullet
attention one period back: L15h7 0.26, L11h1, L23h5, L19h5
Edit A: zero the o_proj columns of the 4 heads
paired n=40: 2.5% vs 0.0%, both 100% cut
failed
Attempt 2: passes 7 to 9
Hypothesis B: layer-23 neurons suppress the next section digit by
~3 nats
n580, n598, n564, n557, n485, n669, n639
Edit B: zero 7 columns in L23, rescale 4 in L22 and L23
paired n=48: 0.0% vs 2.1%, both 100% cut
failed
Attempt 3: passes 10 to 25
Hypothesis C: at the first thinking token </think> is already rank 6
to 8 (logprob -6.9);
4 neurons fire only at that token: L17n537, L18n628, L19n2424,
L20n1235
Edit C: add alpha times the </think> read-out direction to the 4
down_proj columns
alpha = 0.20, 0.20, 0.25, 0.15; 4,083 entries
logprob(</think>) at the first token: -6.9 to -0.11
Confirmation: 7 paired sets of 64 fresh problems, n=448
original 1.1%, edited 28.1%, +26.8 points
122 wrong to right, 2 right to wrong
submitted